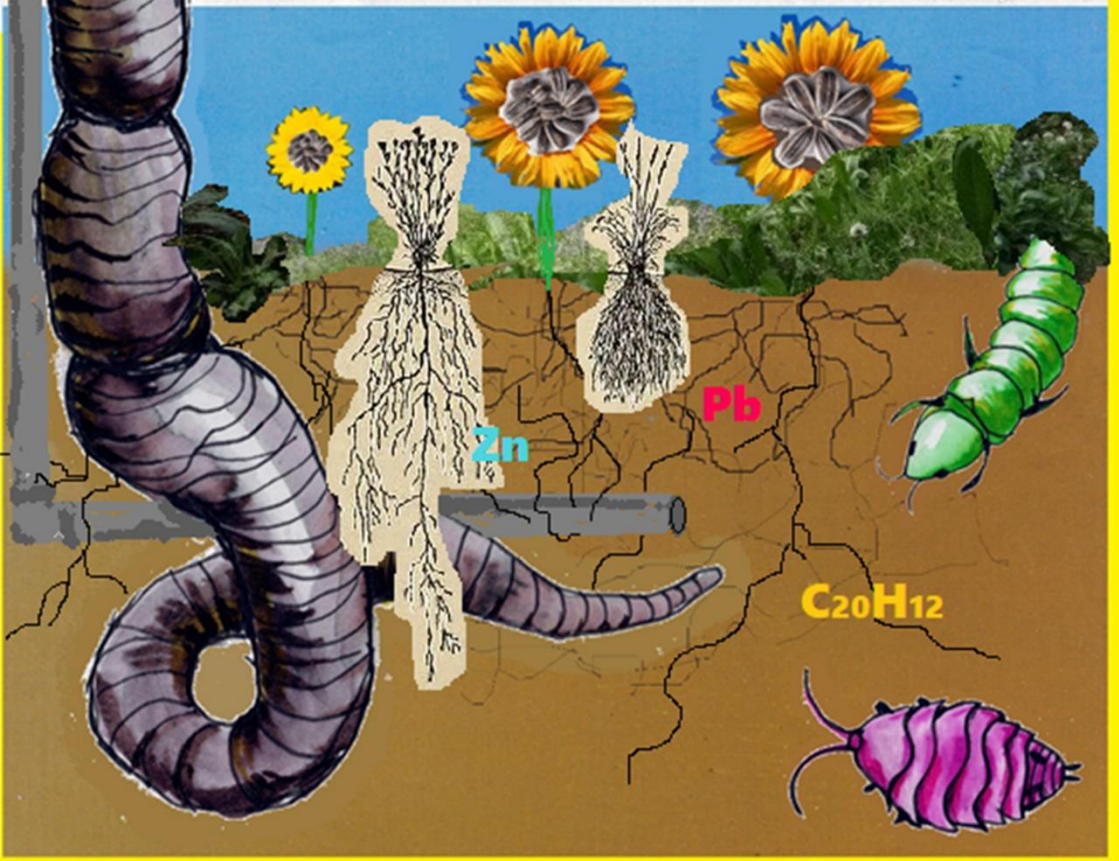
COMPOST MENTIS

1.

A quarantine zine about soil care



seeds ~ stories ~ DIY soil tests ~ rootscapes ~ poems

Written, illustrated and designed by members of Compost Mentis workers co-operative With minibeast illustrations by Nicky Arscott, and additional artwork by Saif Osmani and Mariana Heilmann

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Introduction

Now more than ever seems a good time to be attending to the ground that holds us, with its slowly turning cycles, movements and creeping transformations. For many, whose lives have been put temporarily on hold, or turned upside down, there can be great comfort in turning to the natural world (which we are part of!) and whose processes, and rhythms continue on, from the planetary to the microscopic scale. Whilst nature may be the source of these 'unprecedented times', it is also, as we all know, fragile and in need of care when its systems, communities and cycles are disrupted.

Compost Mentis is concerned with our relationship to soil; how it cares for us and how we can care better for it. The power of composting to produce a material of economic and social value out of "waste" is often overlooked. Processes that we rely on for our food and to deal with our waste are not insulated from shocks, composting can be a way of decentralising these processes and empowering humans and non-humans on a small scale. It connects us with a chaotic and complex world involving the organic and inorganic, oxygen, animals, plants, moisture, smells, colours, touch, emotion and politics.

Delving into the pore spaces and fissures reveals a rich biology of animals, fungi, bacteria and viruses. These pages unearth soil stories, memories and dreams, explore the role of viruses amongst soil communities and question how "dirty" the soil really is (gloves are discretionary). Plant the seeds if you can to help build structure, biodiversity and health in your local soil, or give them to someone who can.

About the authors

Shumaisa Khan follows the meandering path that her zeal for solidarity economics, cultural repair, regenerative systems, and earth-based healing leads her. She sees solutions and possibilities everywhere, and loves facilitating beneficial connections to spread and amplify these. Among her favourite things in life are serendipitous, unplanned interactions – experiences she's devoted to keeping from the brink of extinction.

Linden Katherine McMahon is a writer, performer, and arts & nature connection facilitator. They try to use poetry and stories towards building more just, more sustainable, kinder futures through community and solidarity. They also like baking elaborate desserts, making pots, and reading sci-fi books. http://lindenkatherinemcmahon.org

Olly Edmonds is an ecologist and currently lives in North Devon providing conservation and sustainable land management advice to farmers across Exmoor. He is curious about the interconnectedness within nature, why things are and are not in balance, how spatial scale affects processes and what ecological restoration might look like. He loves bats although would call himself a generalist.

Hari Byles is a community gardener, researcher and compostist. Their focus is on accessibility, sustainability, soil care and alternative sanitation which they attempt to weave together through the building of accessible compost toilets and other composting facilities as part of Compost Mentis. They have been learning about soil microbiology and microscopy with Dr Elaine Ingham, and running workshops to facilitate conversations about soil health and our relationship to microbes.

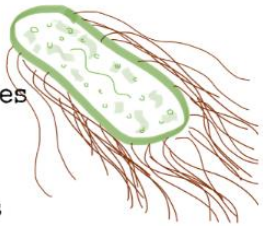
not dirt

shumaisa khan



the little ones know
across time and place,
have always known,
magnetically drawn to the dark aliveness:
soil is a tonic for the soul
science only catching up recently

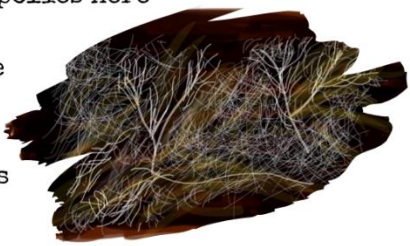
mycobacterium vaccae in garden soil makes us feel good!



microbes and us, an ancient partnership,
a dance boosting serotonin levels
maybe that's why i've avoided gardening gloves
preferring to feel the rich, fecund soil
so teeming with life
a single teaspoon containing more organisms
than there are people on earth
home of possibly the largest organism,
a honey fungus occupying nearly 2400 acres
and at least 2400 years old

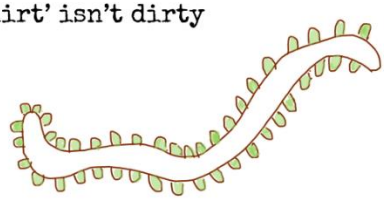


beneath us lies the original world wide web
no polluting servers or corporate monopolies here
a fount of awe and wonder
where materiality and sacredness reside
how many generations past
became the earth we walk upon,
receiving the end of corporial life-ness
a perpetually liminal space
of un-becoming and becoming,
death - inbetweenness - life
nourishing our physical and spiritual body
keeper of our past, present, and future
receiver of grief, channeller of joy
provider of food and medicine
no wonder then
in this time of quarantine
it's demand for soil, not toilet rolls,
that outstrips supply
but in the midst of anxiety and uncertainty
hope, a low-hanging fruit within reach
savor its sweetness
and chuck the scraps in a pile
with the card of spent toilet rolls
marvel as certainty unfolds:
unseen microbial allies
transmuting scraps into dark brown earth
a womb for seeding more new life
so no, 'dirt' isn't dirty



mycellium mat = www





as above, so below

underground flower of violet – this plant spreads its seeds underground as well as above ground!



garlic mustard

a common edible plant

whose root grows in an 's' curve



DIY Soil Structure Tests ~ ~ ~ Hari Byles

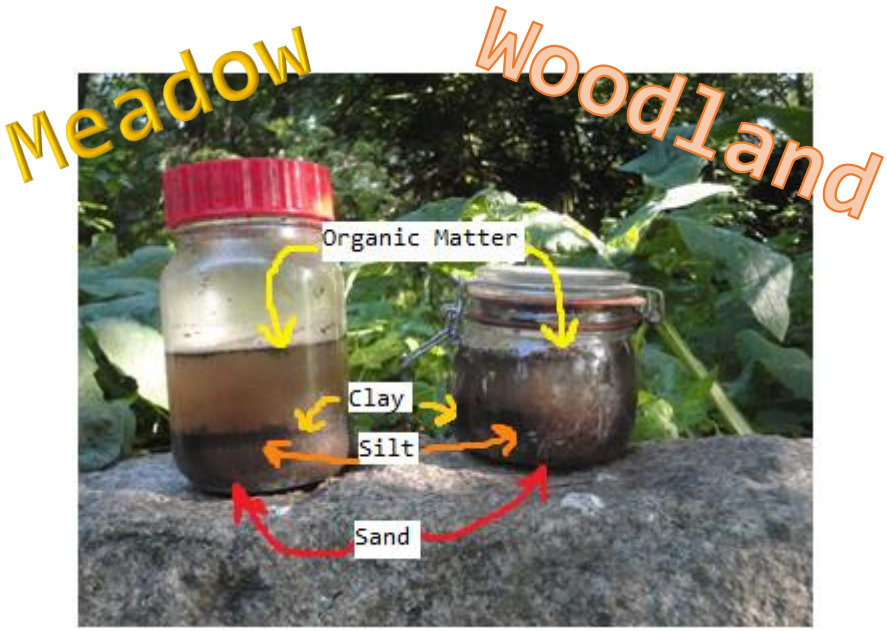


In order to know how to take better care of our soil, we have to find ways of communing with it, to find out how it's doing and what it needs.

One DIY test you can do is to take a soil sample (or several) from an area you want to test. (I dug with a trowel to get some depth, and then put a handful of the loose soil into a jar). Try to avoid scooping up worms, woodlice and other soil friends in your sample.

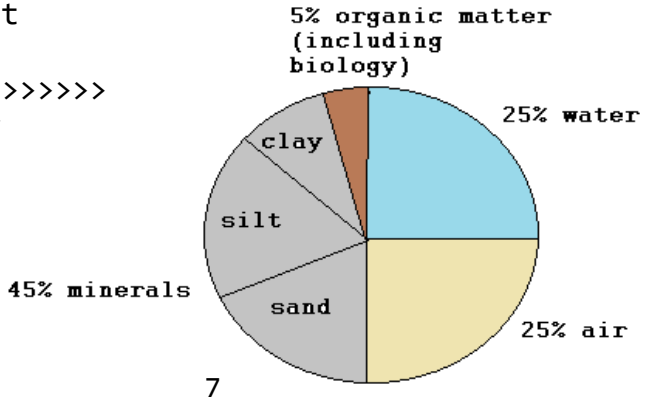
Then, fill the jar with water, shake around, and leave for a couple of hours -> a day.

Once the soil strata has settled you will be able to get a sense of the proportions of sand, silt, clay and organic matter in your soil.



In this example, I compared meadow soil (left) with woodland soil (right). The test shows that the woodland soil has a lot more organic matter and clay compared with the meadow soil which has very little organic matter, about 20% clay, 50% silt and 28% sand. The darker colour of the woodland soil is a further indication that it has a higher organic matter content.

An 'ideal' soil structure might Look like this >>>>>>>>>>>>>>>>>>>> But in reality These patterns Fluctuate all The time.



I did another test comparing samples from different places in the meadow to see if the soil quality varied depending on what was growing:



This test showed pretty similar soil structure in both the comfrey and sweet woodruff areas. With a slightly higher organic matter/ fungal and clay content in the newly imported soil mix.

This is a helpful test to do each year if you are trying to figure out what you might need to add to your soil before the growing season starts.

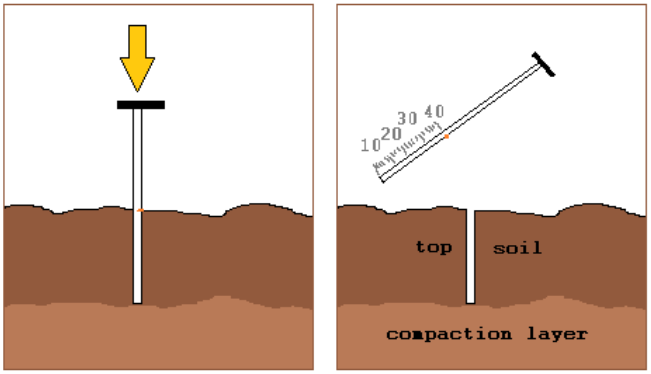
Compaction Test

You can make your own pentrometer to measure top soil depth and compaction using a long thin metal rod with a handle attached to one end.

Other DIY pentrometers include: fingers, fists and pointed sticks.

Measuring top soil depth and compaction is a useful indicator of soil health ~ deeper top soil that is not-compacted allows roots to grow deeper and further, gathering more water and nutrients and supporting more resilient plants.

1) Choose an area you would like to test (preferably a non-stoney area)
2) Push pentrometer into the ground using all your body weight
3) When it feels like you can't go much deeper without applying much more force, mark on the pentrometer where it meets the surface of the ground.
4) Pull the pentrometer out of the soil and measure from your mark to the end.
5) Repeat in several spots to build up a picture of what is happening underground.



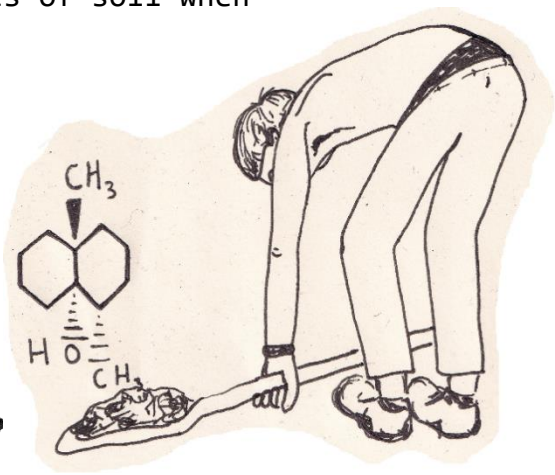
Life and death in the soil
Hari Byles

"In the past scientists suspected that a large DNA-based virus took up residence inside a bacterial cell more than a billion years ago to create the first cell nucleus. If so, then we are all descended from viruses."
(Stupart 2016: 20)



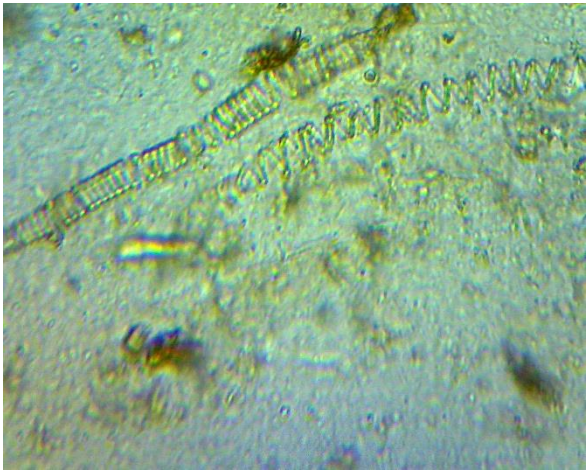
Today (the day after Earth Day and in the midst of the Covid 19 pandemic) I have been tending to some anaerobic compost piles, by turning them and mixing them with carbon materials like toilet roll tubes, used paper towels (from all the handwashing), egg boxes, wood shavings and woodchip. The smell of a compost pile going anaerobic has an unmistakable urgency to it, making me drop all plans (to be working on this zine) and start shovelling, aerating, shredding and mixing. This makes me feel good, reminding me of the medicinal properties of soil when we care for it.

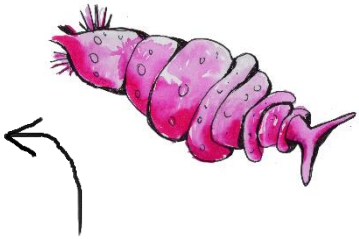
(e.g. Geosmin molecules produced by actinobacteria can have antidepressant affects)



‘Anaerobic’ means low/no oxygen (below 8 parts per million), which creates conditions for different kinds of life to thrive. Anaerobic bacteria, which live in low oxygen conditions, produce acidic glues and compounds, which can have a detrimental effect on plant roots and soil nutrients. In anaerobic conditions essential soluble nutrients like nitrogen, phospherous and sulphur are lost as gases (releasing notable smells of ammonia and rotten egg), so plants aren’t receiving the nutrients they need to grow. I have seen this in gardens with very acidic soil below pH 5; where the roots had become small and clubbed, and the crops were barely growing.

some of the most ancient life forms on the planet thrive in anaerobic environments, as these were dominant 2.4-2.8 billion years ago before oxygen emerged in ouratmosphere.





Anaerobic digestate 400x magnification, spiral shaped organisms are anaerobic

Elaine Ingham describes a smell scale to help determine what is happening on a microscopic scale when soil / compost begins to smell anaerobic:



vinegar/ salad smell = Ascetic acid

sour milk / rancid butter = Butryic acid

vomit ph 2 or lower = Valeric acid

smell of decaying flesh = Putrescine

These smells communicate different things to us, and call us to act in different ways. They are powerful, evocative and sometimes emotional. They are labelled as 'bad' because often they create conditions where disease causing human pathogens can multiply. These are the smells which underlie fears of rot, shit, sickness and death. Lately the entire world has become fixated on one particular pathogenic virus, floating in the air ~ infecting and killing thousands of us. This has made me want to learn about the millions of other kinds of viruses that we are coming into contact with all the time, that live

in our bodies, environments and soils, which connect us symbiotically with other organisms, and are an essential part of our adaptation and survival.

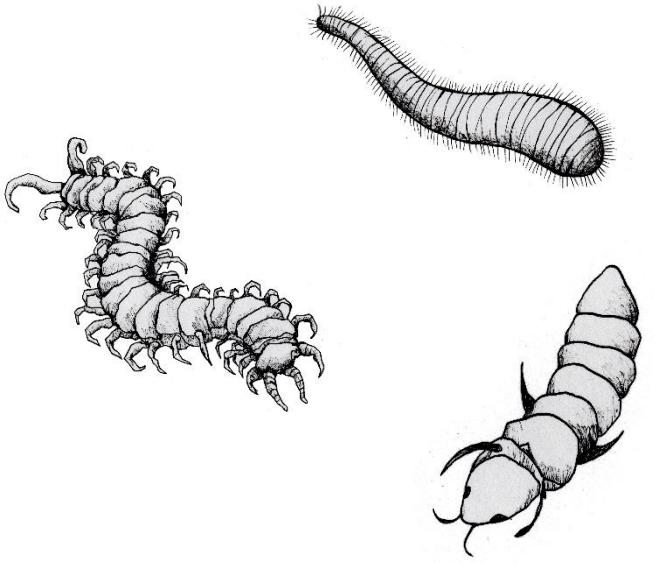
just *a phage* I'm going through?

>>>>>



Viruses break the rules about what it means to be alive, they have been described as 'the undead drivers of microbial life', because they have a huge influence on what lives and what dies in different ecosystems. In soil, there is some recent research which shows that 'phages' (viruses which infect bacterial cells by hacking their DNA and using them to copy their own DNA or RNA, replicating more phages, eventually causing the bacteria to explode!!) play an important role in controlling bacterial populations, and thus releasing the nutrients (particularly carbon) held in bacterial bodies when they die (Kuzyakow and Mason Jones 2018). These nutrients become available to plants through the 'necromass' [I guess this means dead weight?] of the bacteria. In conventional illustrations of the soil food web, viruses (which are about 1000 - 100 times smaller than a single cell bacteria) don't feature, but as almost all

bacteria are infected with various phages, they are an essential part of the knotty complex web of organisms, living, dying, wiggling, feeding, exchanging, eating, reproducing and dancing with each other.



The virus represents an unseen connector of bodies and species, they are the most abundant biological entities on the planet, outnumbering all other biological entities put together. They can infect all living cells including: plants, fungi, animals and bacteria. As well as being pathogenic, they have also been recognised as ‘mutualistic microbial symbionts’ (Roosinck 2015), meaning that they often work in collaboration with their hosts who may in turn be involved in symbiotic relationships with other organisms ~ “These relationships are never black and white; there are costs and benefits, and the players must find a balance. Viruses can be on a continuum between mutualism and antagonism, and where they fall depends on the environment” (Roosinck 2015).

There are lots of examples of mutualistic symbiotic viruses in soil / plant ecology. For instance, viruses which enable plants to survive in extreme temperatures/ drought (look up “panic grass” if you’re interested in this), or avoid attacks from pests (Zucchini yellow mosaic virus). Another nice example of symbiosis in soil is the parasitic relationship between legumes and rhizobium (nitrogen fixing bacteria) which results in large bulging anaerobic nodules forming on the plant’s roots, which house some amazingly complicated chemical processes transforming nitrogen (gas) into a form plants can absorb, also supplying amino acids and proteins to the legume. In white clover, there is a virus (cryptic virus) which acts to suppress this function when there is enough nitrogen available in the soil, saving the plant some effort!

Also, the acidic and alkaline glues/ slime produced by bacteria mentioned above, are sometimes the product of a symbiotic relationship with viruses. Often it is hard to determine where a virus ends and its host begins, because so many of these relationships and collaborations are deeply enmeshed. It is this blurriness and stickiness between where one being ends and another begins, that feels important to acknowledge, during these times of isolation and lockdown. We too can be mutualistic, symbiotic creatures, adding ever more complexity to the pile, becoming part of it as we turn it.

‘Communication happens through involuntary infections and contingent permutations between organisms or substances that attract each other…. Making is always located in mundane interactions and encounters across divergent ecologies. This is the unintentional gift economy of matter and cross-species action. The maker’s words always contaminate each other laterally. Drifting matter.

(Papadopoulos 2014: 641)



Billboard installation by Saif Osmani, Bethnal Green Nature reserve March 2020

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Other stuff:

https://soils.vidacycle.com/soil-tests/

https://en.wikipedia.org/wiki/Virus

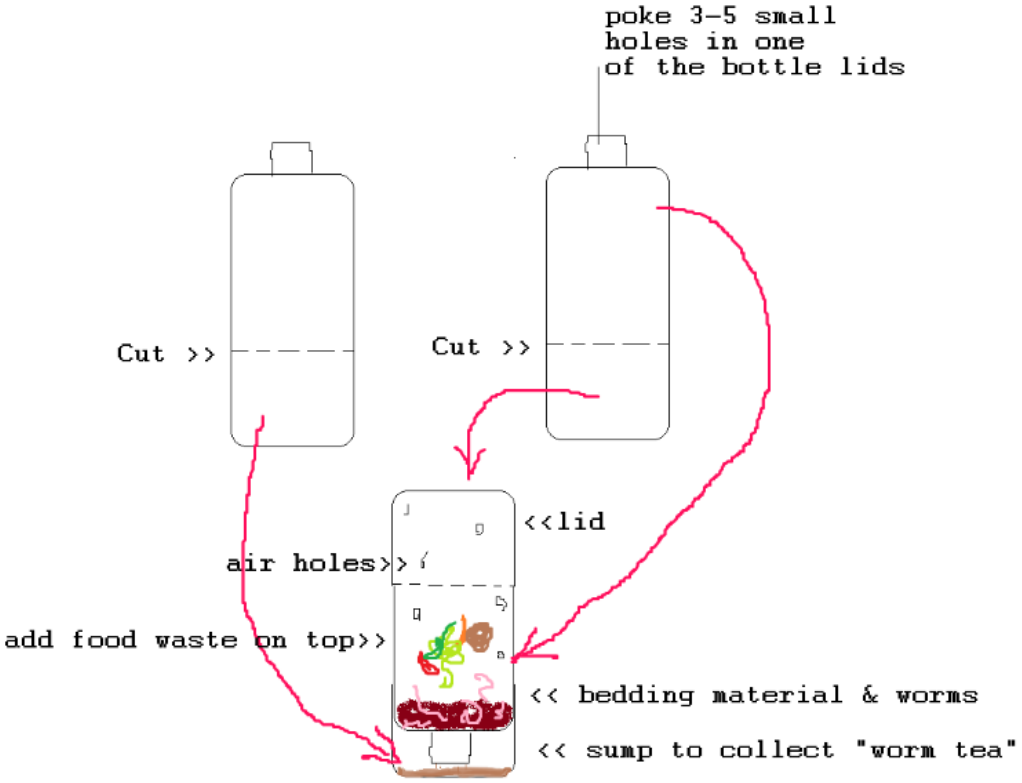
How to make a DIY wormery 'column'
by Mariana Heilmann



Mariana is an artist interested in decay. She came to visit some of the composting systems we manage at Bethnal Green Nature Reserve, to share ideas and knowledge.

She left us with this homemade wormery which we have installed in the compost toilet at the nature reserve, which we helped to build. It is a nice way of demonstrating the composting process to visitors, and also of composting small amounts of kitchen waste.

You will need: 2x clear plastic bottles ~ Scissors ~ Drill or sharp knife ~ Some tiger/ red worms ~ Bedding material like coir, leafmould, or shredded newspaper.



Once in use, cover with dark card / fabric, as worms are sensitive to light

CONNECT

These questions and suggestions are a chance to think about your own personal connection to soil. You can write in the spaces, or use them to have a conversation with someone else :)

A time I remember having my hands in soil is...

Soil makes me think of...

If you can do this safely, take your shoes and socks off and walk barefoot in a grassy spot (keep an eye out for anything sharp!). What can you feel through the soles of your feet?

If you have access to a compost pile or bin, have a look inside and see how many creatures you can find. If there are any you can't identify, see if you can find out what they are...

If you can do this safely in a garden or another quiet spot, lie down so your eyes are at ground level, take some deep breaths, and spend five minutes just watching what's going on there.

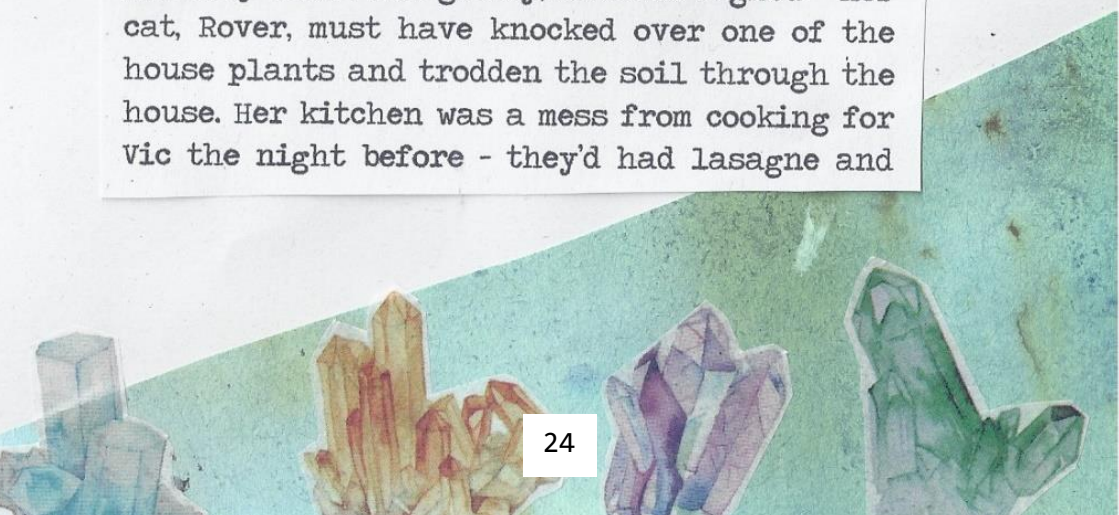
MUD

Maria woke to light across her face where she hadn't closed her curtains properly. They were flimsy anyway, the cheapest the landlord thought he could get away with, and the room was full of soft light. She rolled over, trying to hold onto the feeling from her dream: being held gently by a hand so big she was a speck on the pillow of its palm.

She had been dreaming about the allotment her mum had when Maria was a little kid: just down the road, through a steep park where she'd roll down the hill until her head was spinning. Her dad liked to tell a story (she still didn't know if it was apocryphal) about a dog who tried to pee on one of the trees in that park, but cocked his leg the wrong way and fell down the hill. Through the park and down to the right there was a gate, and through that gate were allotments. Once she was old enough to tell the difference between plants she could pick as many raspberries and peas as she wanted. One year a fox denned under the shed, and she was allowed to stay up late to see the cubs.

In her dream, as in the stories her parents told, her mum had cleared a bed of its potatoes, unearthed them like geodes to be cracked into precious feasts, and then turned the hosepipe on it until it was glittering mud. She gave it to Maria and her brother and the kids of some family friends to wallow and shriek and roll nakedly in for the afternoon - they were completely covered, little wild mud monsters yelling with joy at the playfulness of the earth. In her dream, the brown birthmark on her thigh appeared that day - which couldn't be true, she thought as she woke, could it?

Maria wanted to stay there. But her MacBook called, with its stack of research and writing, and she was desperate for a pee anyway. She stretched and threw off the green leaf-pattered covers, shambled to the bathroom. The hallway floor was gritty, and she sighed - her cat, Rover, must have knocked over one of the house plants and trodden the soil through the house. Her kitchen was a mess from cooking for Vic the night before - they'd had lasagne and



a lot of red wine, danced to Tegan & Sara in the kitchen, and bemoaned the lack of conversational initiative from the people they matched with on OKCupid.

She pieced together a breakfast (banana, coffee, left over garlic bread) and settled down at her desk. Rover demanded her attention as she researched an article about kitchen gadgets, insisting on sitting on her lap as she typed. They found a compromise, and he purred heavily as she lost herself in a world of shiny plastic and chrome. At lunchtime, she watered the plants, forgetting her suspicions about Rover's antics. She plunged her fingers into the soil to check its dampness.

The next night she dreamed of festival mud. Every year at Glastonbury, when she had gone in her early twenties, there were always some people utterly covered in mud. She always wanted to be them, though she wasn't sure why - the wildness, maybe? Play? Wrapping herself in the earth like a blanket? She never did it - you do not know what is in that shit after days

of revelry, and anyway she was utterly self-conscious. In her dream, though, Maria and her friends were both drunk and stoned, giggling and dancing their way through the last night of the festival, and they joked and suggested and eventually dared until Maria and Vic were in a giant mud-puddle and completely covered, falling over each other and laughing uproariously. Maria was watching the sunrise from the Stone Circle with the incessant sound of drums, leaning comfortably against Vic's shoulder, when she woke up to the light through her curtains and builders down the road.

This time, Maria found soil in the angle of the floor and skirting board, drifting from her bedroom window and down the length of the wall towards the door. At first she thought it was some kind of black mould she'd never seen before, and had started steeling herself for an interaction with her landlord before she realised what it was. She swept it up, relieved she wouldn't have to go into battle, distracted by a text from her mum asking what she fancied eating that night when she visited.

That lunchtime, she wandered to the park - something she hadn't done in a while, usually feeling uninspired by the flat expanse of grass. Her brain always buzzed by this point in the day, and the gusty wind and open space cleared her head a bit. She felt a pull, like gravity had suddenly got stronger - she lay down under a tree in the corner of the park and gazed up through the leaves, smelling grass and earth and car exhaust. The ground was dry and firm.

In the next dream her hands were in soil - no gardening gloves - making furrows for seeds and sprinkling them in, the tiny specks sticking to her damp fingers. Her dream-self looked at the soil under her fingernails and smiled at the evidence of her relationship with the earth. Huge raspberry canes drew in water and nutrients around her, dropped heavy ripe berries into her grubby palms. She spotted a worm pulsing away from her furrow, leaned in to watch it. Rover sat on a compost bin nearby, basking in the sun and soaking up the heat from the black plastic. Inside, she knew, all her waste was becoming a gift, and that felt like a gift to her.

Then, half awake, she dreamed of soil working its way twelve floors up her towerblock, all the creatures that live in it - the bacteria and bugs and worms and protozoa and nematodes - working together to climb through the walls and into her bedroom at her call.

When she came fully awake, there were drifts of soil across her bedroom floor, banked like snow against the walls.

She wasn't afraid.

Linden Katherine McMahon

Plants interact in a spectacular number of ways with the soil substrate in which they grow. They shouldn't be thought of as two different components or phases but instead, a complex and tangled network of feedbacks and influences; the soil on the plant and the plant on the soil. Plants can affect the physical, chemical and biological properties and these in turn can affect both the individual plant and the community of plants above ground.

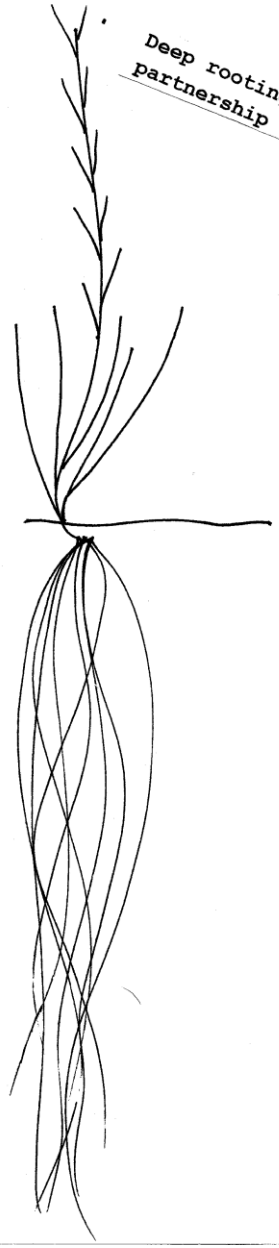
Ultimately this back and forth, pull, squeeze and flow drive more significant change in the whole soil realm, building topsoil, fertility and horizons or layers. The plant community responds and through succession can become more complex (or simple) but often larger, more structured and dominant as it moves towards its climax.

Soils are not inert and like any living organism can react to damage and respond to repair. Sowing diverse plant species with different regenerative properties can help the soil heal with better drainage, movement of air, suppression of disease, space for roots, animals, nutrients and microbes.

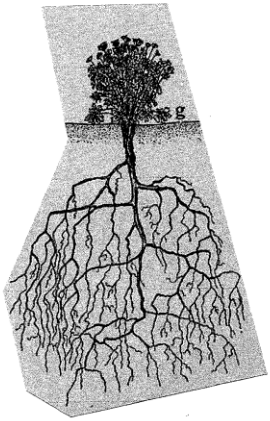
Pioneer species are key to reset and repair the soil and facilitate the slow and gradual development of the soil and plant communities. A stable footing for succession to progress. We may think of these as weeds, but they want to grow; to help the soil and to help us.

Relieve that **compaction**

Deep rooting to break up compaction and engineer the soil in partnership with the earthworms



Ambitious tap roots plunge to the depths of the subsoil to mine for nutrients



Dandelion
Chicory
Ribwort Plantain,
Salad burnet
Yarrow
Borage
Comfrey

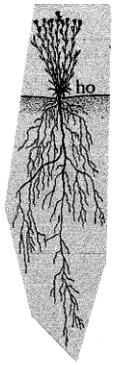
Boost that soil **fertility**

Nettle

Black medick

Curled docks

Vegetation diversity fuels the microbial engine that promotes soil health, fertility, function and resilience



Bird's foot trefoil

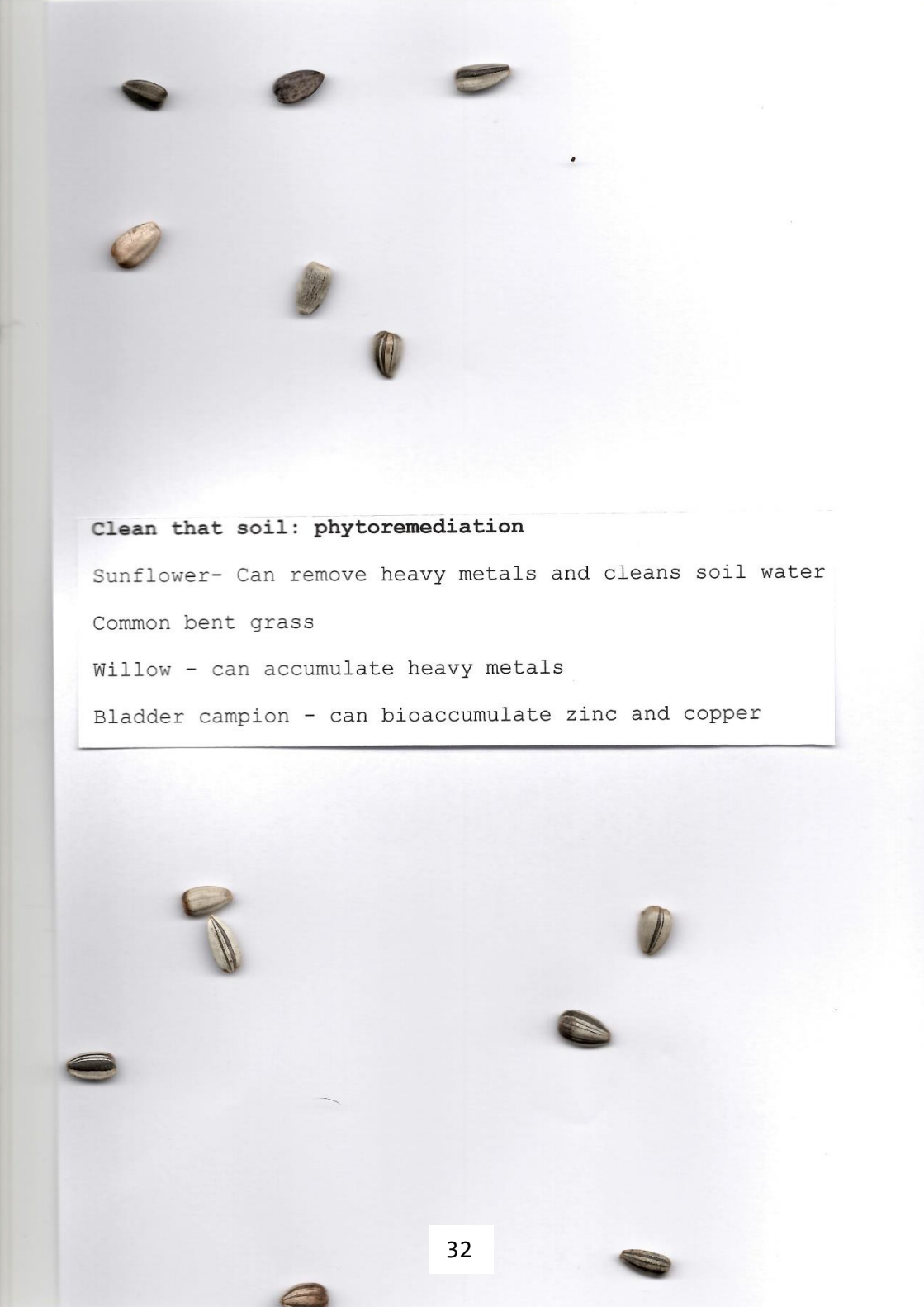
Sainfoin

Red clover (legume) - soil structure, nitrogen, insects

Diverse mycorrhizal communities profit in return for their nutrients

Grasses - fescues and timothy grass

*plant illustrations ~ *Weaver, John E. (John Ernest), 1884-1966; The library of Congress,* courtesy of Wikimedia commons



Clean that soil: phytoremediation

Sunflower- Can remove heavy metals and cleans soil water

Common bent grass

Willow - can accumulate heavy metals

Bladder campion - can bioaccumulate zinc and copper

THE EARTH LOVES US BACK

I want to roll in the earth
and the earth feeds me
muddy baseline alchemy
of light and air through
green bodies of plants
and then smaller beings

we are something queerer
the mud under my nails
like a longed-for tooth-bruise
from a lover uncovered to tell
everyone about us but what lover
could love so big
with the force of a ton of bacteria per acre
what nuance in matter

I love it back as best I can
dig in compost made from the gifts
of my waste becoming gifts of potatoes
sometimes things die because of this love
half of the halved worm becoming
more soil than it already was

what does love mean in the context of death?
we die not for each other but with each other
not a battlefield but something we share
I don't want to make metaphors

out of the deaths of my ancestors
gone from plague and fire before
I was born but they became me
without needing to share blood

just like worms invented hearts to pump it
passed it down through eons
make my present-day body through earth
through vegetables through dinner cooked
by my loves and we cling to each other
soil in the creases of my palms
holding hands watching our leftovers
become beautiful
carrot tops and onion peelings
transubstantiating
into soft dark soil

"The land loves us back. She loves us with beans and tomatoes, with roasting ears and blackberries and birdsongs. By a shower of gifts and a heavy rain of lessons."

\- *Robin Wall Kimmerer, Braiding Sweetgrass*

The seeds in this zine can be sown in pots or into the ground, in spring, summer (with regular watering) or autumn.

If sowing outside, best to find some bare earth or scarify with a rake to open up some bare patches. If the soil is really hard and compacted then consider breaking it up on the surface to provide a good seed bed. Don't overwork the soil or dig too deep - hopefully the plant roots will do a lot of the work to break up compaction lower down. Scatter the seed, tread in lightly and water. New seedlings will need watering regularly to get them going. If sowing the seeds into existing grass, try raking again to break up the sward and provide some gaps for seedlings to germinate. Might be best to hold fire until the autumn if sowing this way, so that the grass doesn't overwhelm the seedlings. Seeds can be stored in a paper bag until needed.

You only need a small pinch of seed per square metre and you can add a bit of sand when you're scattering to help get good coverage.

Sowing into pots is a great way to maximise germination and provide a good burst of foliage and flowers for invertebrates and pollinators. Scatter a small amount of seed and cover with a very fine layer of soil or compost. Water well, and consider planting into the ground once seedlings are more established. The sunflower seed if best sown in spring, make a hole in the ground or a pot about 2cm deep, pop the seed in, and keep watered.

Good luck x